

Is there any anomaly in this product? What category does it belong to, and where is it located?

Single Pass CoT Framework

<think>The sensor exhibits a visible anomaly...This irregular white protrusion suggests a **crack**...The defect is localized in **bottom right**...</think>
<answer>**yes**</answer><type>**cracks**</type>
<bbox> **[400, 600, 420, 650]** </bbox>

Error Classification
↓
Error Localization



Single Pass Agentic Framework

<think>...</think>
<crop>**[100,300, 400,450]**</crop>

Tool Return:



<think>Since the **returned cropped image contains no anomalies**, the product is considered normal</think>
<answer>**no**</answer>

Error Localization ⟶ Task Failure

AirIAD

Spatio-Semantic Analyses

<Spatial think>
The PCB appears mostly intact, but there is a possibly melted or damaged component between the two circular sensors. This region warrants closer inspection to determine.
<Semantic think>
Given the appearance of a distorted or possibly melted component, the most likely defect type is "Melting". I will query this anomaly to verify consistency.

Tool Executor

Tool 1: Perceptive Zoomer
<crop>[390, 300, 565, 390]</crop>
Return:

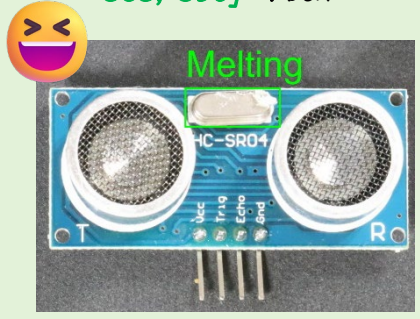


Tool 2: Knowledge Retriever
<query> Melting </query>
Return: Visible melting or thermal deformation on PCB surfaces appears as irregularly shaped, glossy or bubbled regions where...

Agentic Iterative Reasoning

Result Decision

<answer>**yes**</answer>
<type>**Melting**</type>
<box> **[390, 300, 565, 390]** </box>



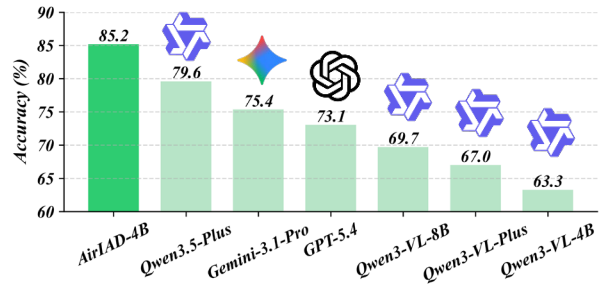
Cross-Verification with Tool Response

<Spatial think>
The zoomed region contains only partial or unclear defect patterns...recall tool to crop...
<Semantic think>
The retrieved knowledge doesn't match the visual evidence... recall tool to query...

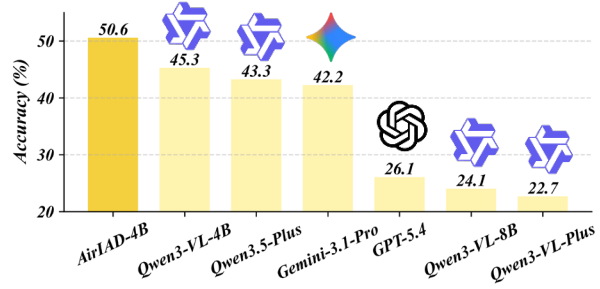
OR

<Spatial think>
The zoomed view **confirms** a defect **aligns** with description...
<Semantic think>
The retrieved knowledge **matches** the visual evidence...

Performance on Anomaly Detection



Performance on End-to-End Detection



Performance across Different Datasets

